

Global CO₂ climate trend: linear or accelerating?

Advanced Statistics for Physics Analysis - Final Project

Fabio Cimino - Daniele Ghezzi

Physics of Data - Università degli Studi di Padova

Atmospheric carbon dioxide (CO₂) is a major driver of the Earth's climate system. Making an accurate quantification of its long-term evolution is essential for understanding anthropogenic climate change. In this study we analyze the Mauna Loa atmospheric CO₂ record from 1958 to 2025 using a Bayesian statistical framework. To evaluate the dynamics of CO₂ accumulation, we compare two competing empirical formulations: a linear model assuming a constant growth rate, and a quadratic model accounting for trend acceleration. Model comparison via leave-one-out cross-validation provides decisive statistical evidence against the constant-growth hypothesis, overwhelmingly favoring the quadratic model. Posterior inference reveals that the annual accumulation rate of atmospheric CO₂ has more than tripled over the observation period, accelerating from approximately 0.77 ppm/yr in 1958 to 2.54 ppm/yr in 2024. Finally, we forecast the underlying atmospheric CO₂ trend to reach approximately 453.0 ppm by 2035 with a 95% credible interval of [452.1, 453.9] ppm. These results highlight the critical necessity of modeling acceleration in atmospheric carbon trends and demonstrate the efficacy of Bayesian methods for environmental forecasting.

INTRODUCTION

Since the Industrial Revolution, human activities have substantially altered the composition of the atmosphere. In particular, the concentration of atmospheric carbon dioxide (CO₂) has increased by approximately 54% relative to the relatively stable levels observed during the 10000 years preceding the mid-eighteenth century. [1] Elevated atmospheric CO₂ concentrations contribute directly to global warming through the greenhouse effect and are also responsible for ocean acidification and increased photosynthesis rates. Consequently, understanding the temporal evolution of atmospheric CO₂ is essential for quantifying anthropogenic impacts on climate, improving projections of future environmental conditions, and informing mitigation strategies.

One of the most important sources of information on long-term atmospheric CO₂ variability are the measurements at the Mauna Loa Observatory in Hawaii. Continuous observations have been collected since 1958, capturing both seasonal fluctuations associated with plant growth and decay and the long-term increase linked to human emissions. A central statistical question is whether the observed long-term increase in atmospheric CO₂ is adequately described by a constant growth rate or whether the growth rate itself increases over time. To investigate this question, two competing trend models are considered. The first is a linear model, which assumes a constant annual increase in the annual-averaged CO₂ concentration. The second is a quadratic model, which introduces an acceleration term and therefore allows the growth rate to vary over time. Although these models are not derived from a physical model of the carbon cycle, they provide a simple and useful framework for characterizing the long-term trend in the observations.

The objective of this study is to analyze the Mauna Loa

CO₂ records using Bayesian inference, compare the performance of the linear and quadratic models, and assess their implications for future atmospheric CO₂ concentrations.

DATASET DESCRIPTION

The analysis is based on the monthly atmospheric CO₂ record from the Mauna Loa Observatory provided by NOAA Global Monitoring Laboratory [2]. The dataset contains observations from March 1958 onward and is publicly available. It provides the monthly mean CO₂ concentration, an associated uncertainty estimate, and the number of observation days contributing to each monthly mean. Because the objective of this study is to model the long-term trend rather than seasonal variability, the monthly observations were aggregated into annual averages. Data collected prior to May 1974 originate from measurements performed at the Scripps Institution of Oceanography. For this portion of the record, information on the number of observation days and uncertainty estimates is not available. Consequently, annual means for these early years were computed as simple arithmetic averages of the available monthly values. From 1975 onward, annual averages were obtained as weighted means of the monthly means, using the number of observation days as weights. The monthly uncertainty estimates were not incorporated into the weighting scheme. These uncertainties are derived from the number of observation days themselves and other factors in a way that is not fully transparent. Moreover, reported measurement uncertainties are of the order of 0.1 ppm and we will find model uncertainty to be much larger, indicating that it arises mostly from structural misspecification (as expected for such empirical models), making more complex

weighting schemes not affect the results. Two months, December 1975 and April 1984, were excluded from the analysis because no direct measurements were reported and the values are derived from interpolation. The first four months of 2026 were also excluded because they do not constitute a complete annual cycle and would therefore introduce seasonal bias into the annual average. The resulting dataset consists of annual average atmospheric CO₂ concentrations spanning 67 years, providing a suitable basis for investigating long-term trends.

Figure 1 and 2 display the monthly and annual averaged atmospheric CO₂ concentration, respectively. A clear long-term increasing trend is visible throughout the entire observation period. The concentration rises from approximately 315.2 ppm in 1958 to 427.5 ppm in 2025, corresponding to an overall increase of approximately 35.6% in 67 years.

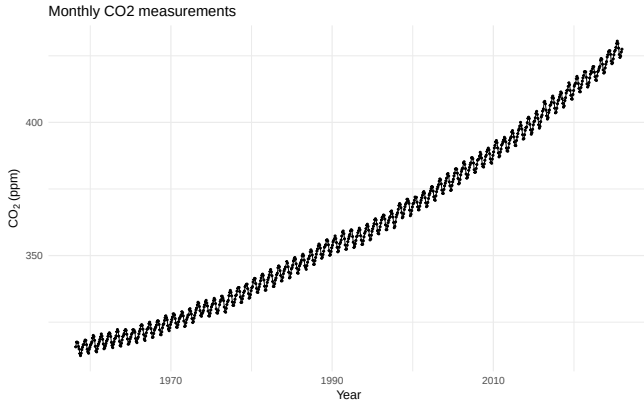


FIG. 1: Monthly atmospheric CO₂ concentration measured at the Mauna Loa Observatory. Data points are connected by lines to aid in visualization.

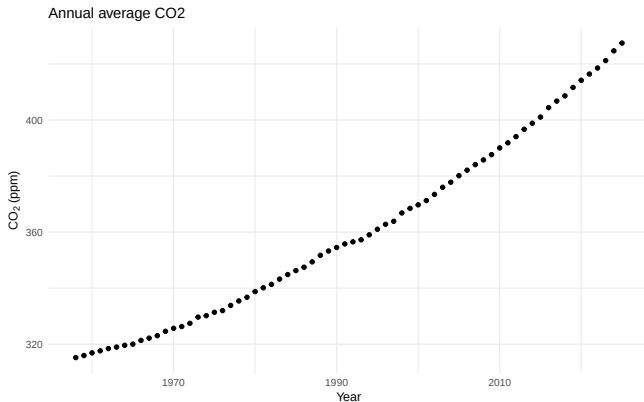


FIG. 2: Annual average atmospheric CO₂ concentration at Mauna Loa Observatory.

METHODS

Trend Models and Likelihood function

Let $y(t)$ denote the observed annual average atmospheric CO₂ concentration in year $t \in T$ where $T = \{1958, 1959, \dots, 2025\}$. We assume that observations arise from an underlying mean trend $\mu(t, \mathbf{b})$ with a Gaussian residual variability of unknown standard deviation σ :

$$y(t) = \mu(t, \mathbf{b}) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2),$$

To describe the evolution of annual-averaged atmospheric CO₂, we consider the linear trend model

$$\mu(t) = b_0 + b_1 t,$$

which assumes a constant annual increase in atmospheric CO₂, and the quadratic trend model

$$\mu(t) = b_0 + b_1 t + b_2 t^2,$$

which allows the rate of increase itself to increase at a constant rate. We assume residuals to be conditionally independent given the parameters $\mathbf{b}$ and σ . This implies conditional independence of the observations $D = (y_t)_{t \in T}$, and thus the likelihood factorizes as

$$p(D|\mathbf{b}, \sigma) = \prod_{t \in T} \mathcal{N}(y_t; \mu(t, \mathbf{b}), \sigma^2).$$

The objective is to determine which of the two trend models provides a better description of the observed data and to use the preferred model to generate forecasts for the next 10 years $t \in \{2026, 2027, \dots, 2035\}$ within the Bayesian statistical framework.

Posterior computation

For a given model, the posterior distribution can be obtained through Bayes' theorem by multiplying the likelihood by the prior distribution (which will be discussed later):

$$p(\mathbf{b}, \sigma|D) = \frac{p(D|\mathbf{b}, \sigma)p(\mathbf{b}, \sigma)}{p(D)} \propto p(D|\mathbf{b}, \sigma)p(\mathbf{b}, \sigma).$$

However, the posterior distributions associated with both models are not available in closed form and therefore must be approximated numerically. All Bayesian computations are performed using Stan, a probabilistic programming language designed for Bayesian inference and statistical modeling, via its R interface `cmdstanr`. Posterior sampling is carried out using Hamiltonian Monte Carlo (HMC) with the No-U-Turn Sampler (NUTS) [3], Stan's adaptive implementation of HMC. Unlike conventional random-walk Markov chain Monte Carlo (MCMC)

algorithms, HMC exploits gradient information from the posterior distribution to generate efficient proposals and explore high-dimensional parameter spaces. The NUTS algorithm automatically determines an appropriate trajectory length during sampling, reducing the need for manual tuning and improving computational efficiency.

For each model, four independent chains are initialized from dispersed starting values. Each chain is run for 2000 warm-up iterations followed by 10,000 sampling iterations, yielding a total of 40,000 posterior draws per model. The warm-up phase is used for adaptation of the HMC tuning parameters and to reach stationarity of the chain; it is discarded from the final analysis. Posterior summaries and predictive quantities are computed from the retained samples.

Data standardization

Polynomial models can mix slowly because coefficients compensate each other. In particular, a quadratic model creates long and tilted posterior contours, resulting in inefficient exploration of the posterior distribution. A remedy is to center and scale the input variable:

$$s = \frac{t - \bar{t}}{\sigma_t}, \quad (1)$$

where $\bar{t} = 1991.5$ and $\sigma_t = 19.77$ are the mean and standard deviation of the observation years, respectively. Given our data, s ranges between -1.7 and 1.7 . Fitting both the linear and quadratic models using the standardized variable s leads to:

$$\begin{aligned} y &= \tilde{b}_0 + \tilde{b}_1 s + \varepsilon, \\ y &= \tilde{b}_0 + \tilde{b}_1 s + \tilde{b}_2 s^2 + \varepsilon, \end{aligned}$$

which substantially reduces posterior correlations among the regression coefficients, ensuring highly efficient sampling.

Priors

Under this standardization procedure, the intercept $\tilde{b}_0$ approximately represents the CO₂ concentration at the middle of the observation period ($s = 0$). To reflect prior ignorance while restricting the parameter to physically plausible values, a uniform prior is assigned:

$$\tilde{b}_0 \sim \text{Uniform}(0, 1000) \text{ ppm.}$$

The slope parameter $\tilde{b}_1$ determines the average increase in atmospheric CO₂ per standard deviation of time σ_t . Rather than assigning a prior directly to the slope, the angle

$$\alpha = \arctan(\tilde{b}_1)$$

is introduced and assigned a uniform prior:

$$\alpha \sim \text{Uniform}\left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

This choice corresponds to a prior that is invariant with respect to orientation in the (s, y) plane and avoids placing arbitrary upper bounds on the growth rate. The resulting induced prior on $\tilde{b}_1$ is weakly informative.

Finally, the noise scale is assigned a prior that is uniform on the logarithmic scale:

$$\log \sigma \sim \text{Uniform}(-10, 5),$$

which corresponds to a physically reasonable, scale-invariant prior for $\sigma \in [e^{-10}, e^5] \approx [5 \cdot 10^{-5}, 150]$ ppm.

For the quadratic model, the priors for $\tilde{b}_0$, $\tilde{b}_1$, and σ are unchanged. We adopt a broad uniform prior on the acceleration coefficient:

$$\tilde{b}_2 \sim \text{Uniform}(-500, 500) \text{ ppm.}$$

Since the standardized time s varies approximately between -1.7 and $+1.7$, the maximum absolute contribution of the quadratic term $\tilde{b}_2 s^2$ is bounded by roughly 1500 ppm. This vastly exceeds the total variation of CO₂ that we can expect across the historical record, ensuring that the prior does not artificially constrain the posterior distribution.

Model diagnostics

The reliability of the posterior inference is assessed using standard diagnostic metrics for MCMC algorithms. The convergence of the independent chains to the target posterior distribution is evaluated using the Gelman–Rubin convergence diagnostic $\hat{R}$. This statistic compares the between-chain variance to the within-chain variance, and a value close to one indicates that the chains have successfully mixed.

Sampling efficiency is quantified via the effective sample size (ESS), which estimates the number of independent draws equivalent to the collected MCMC samples, which are inherently autocorrelated. Following the Stan implementation, both bulk ESS and tail ESS are monitored. Bulk ESS evaluates the sampling efficiency in the main body of the distribution, ensuring reliable estimates of the posterior mean and median. Tail ESS evaluates the exploration of the distribution's extremes, ensuring accurate computation of variance and credible intervals.

All other Stan diagnostics are also monitored. First, Stan checks for divergent transitions, which would indicate regions of high curvature that the sampler failed to resolve. It also checks for a satisfactory value of the energy Bayesian fraction of missing information (EBFMI), which evaluates the efficiency of the sampler's

momentum by comparing the variance of energy transitions between successive iterations to the overall variance of the energy. Finally, it verifies that no transitions exceeded the maximum allowed treedepth, confirming the efficiency of the NUTS trajectory exploration and the absence of problematic posterior geometries.

Model comparison

Model comparison between the linear and quadratic models is performed using the Expected Log Predictive Density (ELPD) via the leave-one-out cross-validation (LOO-CV) [4]. LOO evaluates the predictive accuracy of a model by fitting it N separate times. In each iteration i , a single data point y_i is left out, the model is trained on the remaining $N - 1$ points (denoted as y_{-i}), and this newly trained model is used to predict the probability of the omitted point. The ELPD aggregates these results by summing the expected log-predictive probabilities for all N omitted points: $\text{ELPD} = \sum_{i=1}^N \log(p(y_i | y_{-i}))$. Since refitting a Bayesian MCMC model N times is computationally expensive, the `loo` package in R approximates this exact process using Pareto Smoothed Importance Sampling (PSIS) applied directly to the point-wise log-likelihoods. To determine the preferred model, we compute the difference in their expected log predictive densities ΔELPD . The superiority of a model is considered statistically significant if the absolute value of ΔELPD is substantially larger than its estimated standard error.

Posterior Predictive Checks

To evaluate the goodness of the fitted models and their generative performance, we employ Posterior Predictive Checks (PPCs), which compare the true data distribution with datasets generated from the model. Practically, for each HMC draw of the parameters $\tilde{\theta}^{(i)}$, we simulate a new set of observations $\mathbf{y}^{rep.(i)}$ for each $t \in T$ from the corresponding Gaussian likelihood. By overlaying the kernel density estimate of the empirical data distribution with those of the replicated datasets, we can visually diagnose systematic discrepancies. A good model should produce a family of simulated density curves that envelops the empirical density, whereas models with missing structure will manifest with regions where the observed data clearly deviates from the predictive one.

Prediction

A primary objective of this study is to forecast future atmospheric CO_2 concentrations. In the Bayesian framework, predictions are generated by simulating from the posterior predictive distribution.

Let $\tilde{y}$ denote an unobserved future measurement at standardized time $\tilde{s}$, and let $\mathcal{D}$ denote the observed data. The posterior predictive distribution is given by the integral of the data-generating model over the joint posterior distribution of the parameters $\tilde{\theta}$:

$$p(\tilde{y}|\mathcal{D}) = \int p(\tilde{y}|\tilde{\theta})p(\tilde{\theta}|\mathcal{D})d\tilde{\theta}.$$

In practice, this integral is approximated using the drawn MCMC samples. For each posterior sample of the regression coefficients and the residual scale σ , a simulated future observation is generated from the corresponding likelihood. This naturally encompasses all the uncertainties arising from residual variability and parameter uncertainty. Therefore, to properly disentangle these sources of uncertainty, we evaluate two distinct intervals:

- **Credible band for the regression curve:** This describes the uncertainty in the noise-free mean function, $\mu(\tilde{s}; \tilde{\theta})$. In practice, given the posterior samples $\tilde{\theta}^{(i)}$ obtained via HMC, one computes the expected value $\mu^{(i)}(\tilde{s}) = \mu(\tilde{s}; \tilde{\theta}^{(i)})$ and extracts the 2.5% and 97.5% quantiles for a 95% credible interval.
- **Posterior predictive band:** This describes the total uncertainty regarding a future measurement of the value $\tilde{y}$. It includes both the uncertainty in the mean function and the observational noise. For a Gaussian noise model, predictive draws are generated by first taking the posterior samples $\tilde{\theta}^{(i)}$ and $\sigma^{(i)}$, and then drawing a new observation $\tilde{y}^{(i)} \sim \mathcal{N}(\mu(\tilde{s}; \tilde{\theta}^{(i)}), (\sigma^{(i)})^2)$. Because it accounts for residual data variance, the 95% predictive band is inherently wider than the credible band for the mean curve.

RESULTS

MCMC Diagnostics

For both the linear and quadratic models, no issues were detected in the sampling procedure. Stan reported satisfactory treedepth for all transitions, satisfactory E-BFMI and no divergent transitions. Trace plots and autocorrelation functions (shown in Appendix) show well-mixed chains with rapidly decaying autocorrelation. All parameters in both models exhibited $\hat{R}$ values of 1.00, indicating convergence across chains. Effective sample sizes were large relative to the total number of retained draws (40,000) as reported in Table I. These values indicate low autocorrelation and highly efficient exploration of the posterior distribution.

Model	Parameter	Bulk ESS	Tail ESS
Linear	$\tilde{b}_0$	45619	30746
	$\tilde{b}_1$	29964	26680
	σ	27928	24127
Quadratic	$\tilde{b}_0$	39102	25882
	$\tilde{b}_1$	16999	17435
	$\tilde{b}_2$	40116	29901
	σ	14465	15485

TABLE I: Effective sample sizes for the linear and quadratic models. The bulk ESS for $\tilde{b}_0$ slightly exceeds the total number of draws, which can occur under mild negative autocorrelation or due to finite-sample estimation effects in ESS calculations.

Posterior pair plots (Figure 3 and Figure 4) and correlation values reported in the figures show no significant correlation among parameters except for a small correlation between $\tilde{b}_0$ and $\tilde{b}_2$ in the quadratic model. This anti-correlation is a known structural feature because the standardized variable s is centered around zero while the quadratic term s^2 is strictly non-negative. To maintain the regression curve close to the observed data, any proposed increase in the quadratic curvature ($\delta\tilde{b}_2 > 0$) must be dynamically compensated by a downward shift of the intercept ($\delta\tilde{b}_0 < 0$), with the constraint $\delta\tilde{b}_0 \approx -\delta\tilde{b}_2 s^2$. Despite this structural coupling, the temporal standardization described in the Methods section successfully resolved the typical correlation which arise in polynomial models. Overall, all diagnostic checks indicate that posterior inference can be considered reliable for both models.

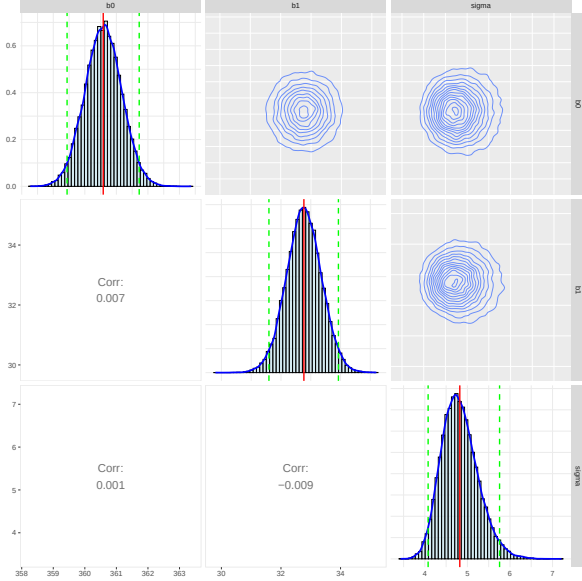


FIG. 3: Posterior pair plots and correlation values for the linear model after standardization.

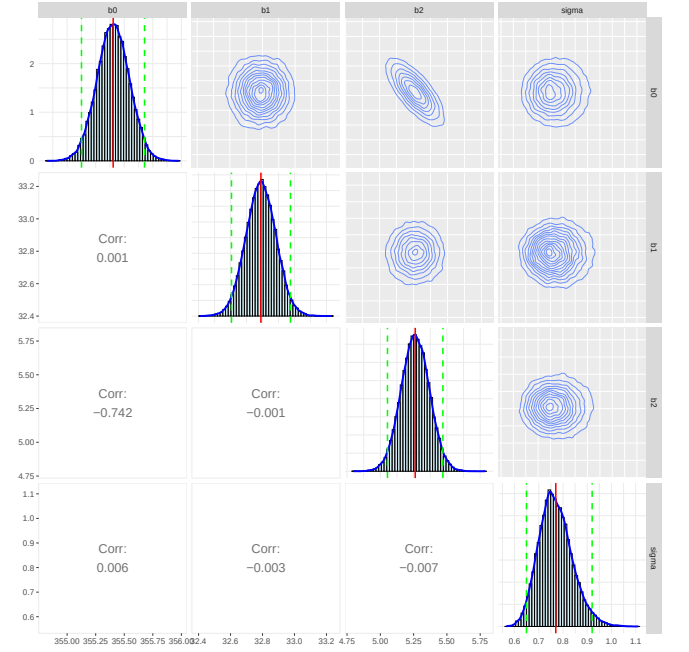


FIG. 4: Posterior pair plots and correlation values for the quadratic model after standardization.

Linear Model Results

Posterior summaries for the linear model parameters are reported in Table II.

Parameter	Mean	MAP	SD	95% CI
$\tilde{b}_0$ (ppm)	361.0	360.6	0.6	[359.4, 361.7]
$\tilde{b}_1$ (ppm)	32.8	32.8	0.6	[31.6, 33.9]
σ (ppm)	4.82	4.70	0.43	[4.08, 5.76]

TABLE II: Posterior summaries for the linear model.

Figure 5 shows the observed annual CO_2 concentrations together with the posterior mean fit and associated uncertainty bands. The red region represents the 95% credible interval of the mean prediction, while the gray region corresponds to the 95% posterior predictive interval. Figure 6 shows the same plot with the mean curve subtracted for better visualization. Although the regression curves and uncertainty bands are plotted as continuous functions for visual clarity, it is important to remark that $y(t)$ represents the average CO_2 concentration in year t which is inherently discrete.

Although the model captures the overall upward trend, systematic deviations are evident. This pattern suggests that the assumption of a constant growth rate is insufficient to describe the long-term evolution of atmospheric CO_2 . The estimated concentration increase, which is

constant in the linear model, is

$$\mu(t+1) - \mu(t) = \frac{\tilde{b}_1}{\sigma_t}.$$

The shape of the posterior is the same as that of $\tilde{b}_1$ shown in Figure 3; its mean is 1.66 ppm/yr and the 95% CI is [1.60, 1.71] ppm/yr.

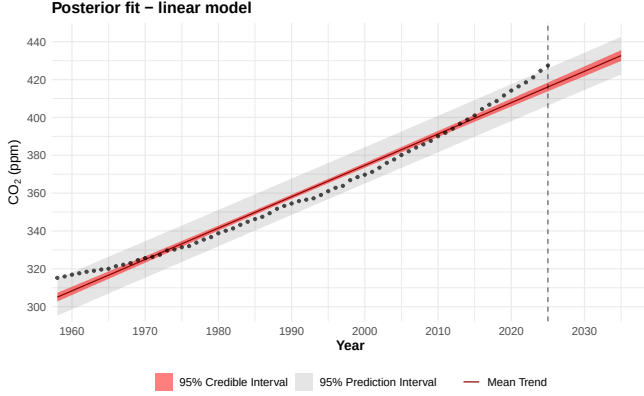


FIG. 5: Linear model fit to annual CO₂ concentrations. The solid line shows the posterior mean, the narrow red band represents the 95% credible interval of the mean, and the wider gray band represents the 95% posterior predictive interval.

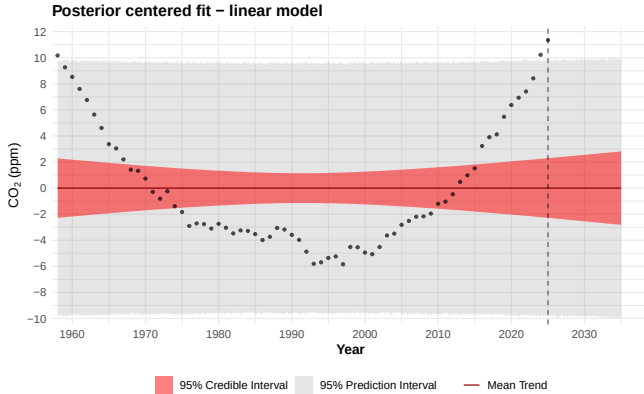


FIG. 6: Centered version of plot in Figure 5, obtained by subtracting the mean curve.

Quadratic Model Results

Posterior summaries for the quadratic model are presented in Table III.

Parameter	Mean	MAP	SD	95% Credible Interval
$\tilde{b}_0$ (ppm)	355.0	355.4	0.1	[355.1, 355.7]
$\tilde{b}_1$ (ppm)	32.8	32.8	0.1	[32.6, 33.0]
$\tilde{b}_2$ (ppm)	5.26	5.26	0.11	[5.05, 5.47]
σ (ppm)	0.77	0.75	0.07	[0.65, 0.92]

TABLE III: Posterior summaries for the quadratic model.

Figure 7 shows the observed data together with the posterior mean, 95% credible interval of the mean, and 95% posterior predictive interval. Figure 8 shows the same plot with the mean curve subtracted for better visualization.

Compared to the linear model, the quadratic specification tracks both the long-term trend and the increasing curvature of the data more accurately. The residual variability is substantially smaller than that of the linear model, as reflected by the posterior mean estimate $\sigma = 0.77$ ppm. For the linear model the mean was $\sigma = 4.82$ ppm. The credibly positive quadratic coefficient $\tilde{b}_2$ indicates clear evidence of acceleration in atmospheric CO₂ growth over time. The distribution pattern of data points around the mean curve still suggests, as expected, that the quadratic model does not capture all the features of the dataset.

The CO₂ concentration growth is not constant in the quadratic model, and it is given by:

$$\Delta\mu(t) = \mu(t+1) - \mu(t) = \frac{\tilde{b}_1}{\sigma_t} + \tilde{b}_2 \frac{1 + 2(t - \bar{t})}{\sigma_t^2}$$

Using the MCMC samples of the parameters $\tilde{b}_1, \tilde{b}_2$, the distribution of the growth rate can be estimated for a given t . In 1958 the mean is equal to 0.77 ppm/yr (CI [0.73, 0.81] ppm/yr), while it is 2.54 ppm/yr (CI [2.51, 2.58] ppm/yr) in 2024. Both concentration growth rate distributions are symmetric, unimodal, and approximately Gaussian; these are shown in the Appendix. The growth rate itself increases annually by a constant acceleration:

$$\Delta\mu(t+1) - \Delta\mu(t) = \frac{2\tilde{b}_2}{\sigma_t^2}$$

of mean 0.027 ppm/yr² and CI [0.026, 0.028] ppm/yr², and its distribution is a rescaled version of the $\tilde{b}_2$ distribution shown in Figure 4.

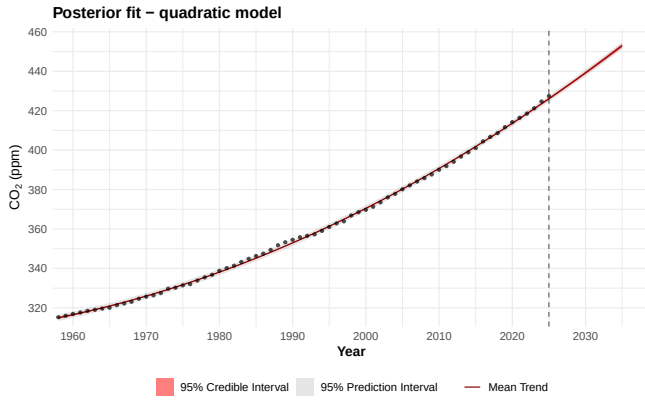


FIG. 7: Quadratic model fit to annual CO_2 concentrations. The solid line shows the posterior mean, the inner red band the 95% credible interval of the mean, and the outer grey band the 95% posterior predictive interval.

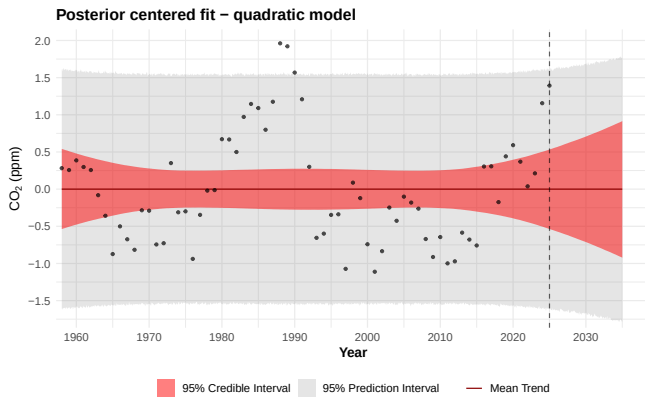


FIG. 8: Centered version of plot in Figure 7, obtained by subtracting the mean curve.

Model Comparison

Model comparison was performed using expected log predictive density (ELPD) via leave-one-out cross-validation (LOO-CV). Relative to the quadratic specification, the linear model exhibited a decrease in expected log predictive density of 124.4 with a standard error of 7.7, indicating substantially poorer out-of-sample predictive performance. The magnitude of this difference provides decisive evidence that a constant growth-rate model is inadequate and that acceleration is an essential feature of the long-term atmospheric CO_2 trajectory.

Furthermore, this superior fit is directly reflected in the posterior estimates for the residual noise σ . The 95% credible interval for σ dropped from $[4.08, 5.76]$ ppm in the linear model to $[0.65, 0.92]$ ppm in the quadratic model. This significant reduction demonstrates the quadratic model's ability to effectively capture the structural variation in the data without resorting to high noise values.

Posterior Predictive Checks

The generative performance of the linear and quadratic models was evaluated using Posterior Predictive Checks (PPCs). Figure 9 and Figure 10 show the density distribution of the observed CO_2 concentrations (dark line) compared to 100 replicated datasets (light lines) generated by the models. The true data has a clear peak at lower CO_2 values, representing the slower growth in the early years of the record, and a long tail capturing the rapid increase in recent decades.

For the linear model (Figure 9), the simulated datasets fail to reproduce this shape. The generated curves are too flat and completely miss the peak of the observed data. This visual check confirms that this model lacks the necessary structural acceleration to describe the real dataset.

On the other hand, the quadratic model (Figure 10) provides a much better fit. The empirical data line is largely enveloped by the simulated density curves, successfully reproducing the peak and the overall shape of the distribution. Although the overlap is not perfect, the generated datasets closely resemble the real observations. This confirms that the acceleration term is required to properly describe the overall atmospheric CO_2 trend.

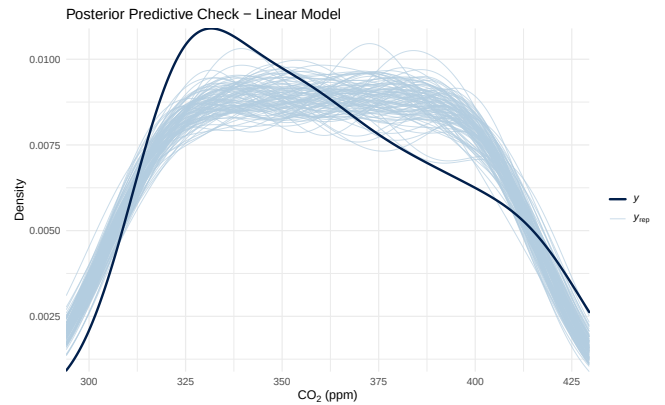


FIG. 9: Posterior predictive check for the linear model. The dark line represents the true CO_2 density, while the light lines are 100 simulated datasets.

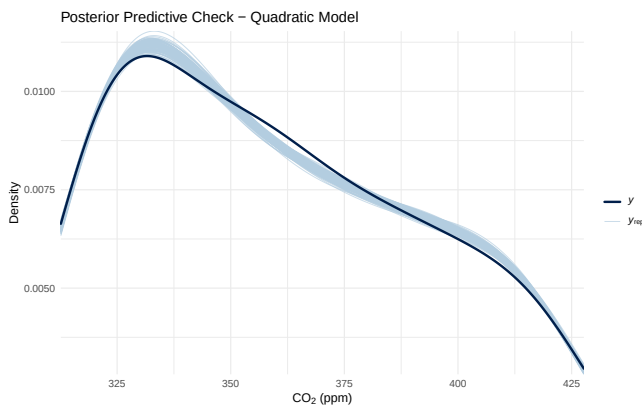


FIG. 10: Posterior predictive check for the quadratic model. The simulated datasets (light lines) largely envelop the true data density (dark line), showing a significant improvement over the linear assumption.

Forecasting

Using the quadratic model, posterior forecasts of the mean function were generated for the next decade (Table IV). We focus on the posterior of the mean rather than the posterior of an observation, as our objective is to quantify the long-term structural climate trend rather than predicting future measurements, which are subject to noise. The model predicts continued growth in atmospheric CO_2 concentrations, reaching approximately 453.0 ppm by 2035, as shown also in Figure 7. Forecast uncertainty increases gradually with the prediction horizon due to parameters uncertainties, as shown also by the red band in Figure 8. Note however that these intervals quantify uncertainty conditional on the quadratic model and should not be interpreted as comprehensive uncertainties on forecasts of future CO_2 concentrations. The full posterior for the year 2035 is shown in Figure 11. Posterior mean plots for all future years have been inspected and they all present a symmetric, unimodal, Gaussian-like distribution.

Year	Mean forecast (ppm)	95% CI (ppm)
2026	428.6	[428.1, 429.2]
2027	431.2	[430.6, 431.8]
2028	433.8	[433.2, 434.5]
2029	436.5	[435.8, 437.2]
2030	439.2	[438.5, 439.9]
2031	441.9	[441.2, 442.7]
2032	444.6	[443.9, 445.4]
2033	447.4	[446.6, 448.2]
2034	450.2	[449.3, 451.1]
2035	453.0	[452.1, 453.9]

TABLE IV: Posterior forecasts of the mean function from the quadratic model: means and credible intervals for each year in the next decade.

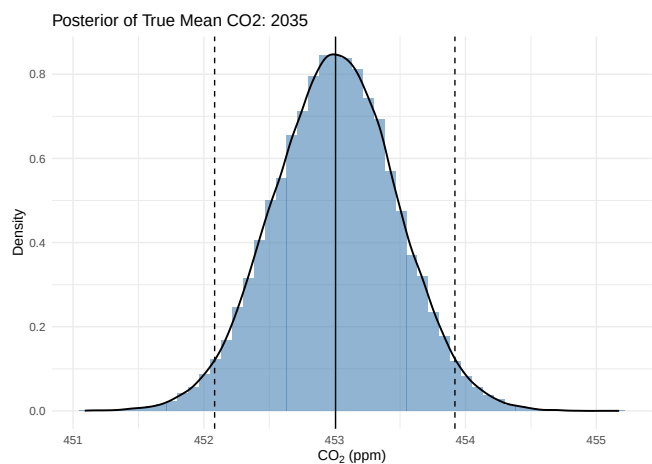


FIG. 11: Posterior predictive distribution for the mean atmospheric CO_2 concentration in the year 2035.

DISCUSSION

Physical interpretation

The results provide clear statistical evidence that atmospheric CO_2 concentrations have not increased at a constant rate during the observational record. Instead, the rate of increase itself has accelerated. The posterior estimate of the acceleration parameter of approximately 0.027 ppm/yr^2 implies that the annual growth rate rose from approximately 0.77 ppm/yr between 1958 and 1959 to approximately 2.54 ppm/yr between 2024 and 2025.

This finding is consistent with the historical increase in anthropogenic greenhouse-gas emissions and with independent analyses of global carbon-cycle observations. Although the present model is purely statistical and does not explicitly represent carbon-cycle processes, the inferred acceleration reflects a persistent long-term change in the atmospheric accumulation of carbon dioxide.

Limitations

Several limitations should be acknowledged. First, the analysis is based on measurements from a single monitoring station. Although the Mauna Loa record is widely regarded as representative of global atmospheric trends, local influences cannot be excluded a priori.

Second, the models considered are empirical trend models rather than mechanistic representations of the carbon cycle. Consequently, the estimated model parameters do not map directly to physical quantities.

Third, forecasts assume that the historical trend continues into the future. Future concentrations may differ from these projections if major changes occur in global emissions, land-use practices, carbon-capture technologies, or climate policy. Therefore, the forecasts should be interpreted as extrapolations of the historical trend rather than predictions based on emissions scenarios.

Future work

Future analyses could incorporate seasonal variability directly through hierarchical models, combine observations from multiple monitoring stations, or implement physical carbon-cycle models. Such approaches would provide a more complete representation of both short-term variability and long-term climate dynamics.

CONCLUSION

Bayesian regression models were used to investigate the long-term evolution of atmospheric CO_2 concentrations measured at the Mauna Loa Observatory. Comparison of linear and quadratic trend models demonstrated that the observed record cannot be adequately described by a constant growth rate. Instead, strong evidence was found for acceleration in atmospheric CO_2 accumulation. The quadratic model yielded an estimated acceleration of approximately 0.027 ppm/yr^2 (CI $[0.026, 0.028] \text{ ppm/yr}^2$). The inferred annual growth rate increased from approximately 0.77 ppm/yr (CI $[0.73, 0.81] \text{ ppm/yr}$) in 1958 to approximately 2.54 ppm/yr (CI $[2.51, 2.58] \text{ ppm/yr}$) in 2024. Forecasts based on the quadratic model suggest that atmospheric CO_2 concentrations will continue to rise throughout the next decade, reaching approximately 453 ppm by 2035 if the historical trend persists.

APPENDIX

Trace and Autocorrelation Plots

This section reports the diagnostic plots for the Hamiltonian Monte Carlo (HMC) sampling procedure. For both the linear and quadratic models, the trace plots

demonstrate excellent mixing and stationarity. The four independent chains rapidly converge to the same regions and explore the whole parameter space.

Furthermore, the autocorrelation function (ACF) plots show a rapid decay to zero within a few lags for all parameters. This rapid drop indicates low autocorrelation between consecutive MCMC samples, confirming the high sampling efficiency and large effective sample sizes.

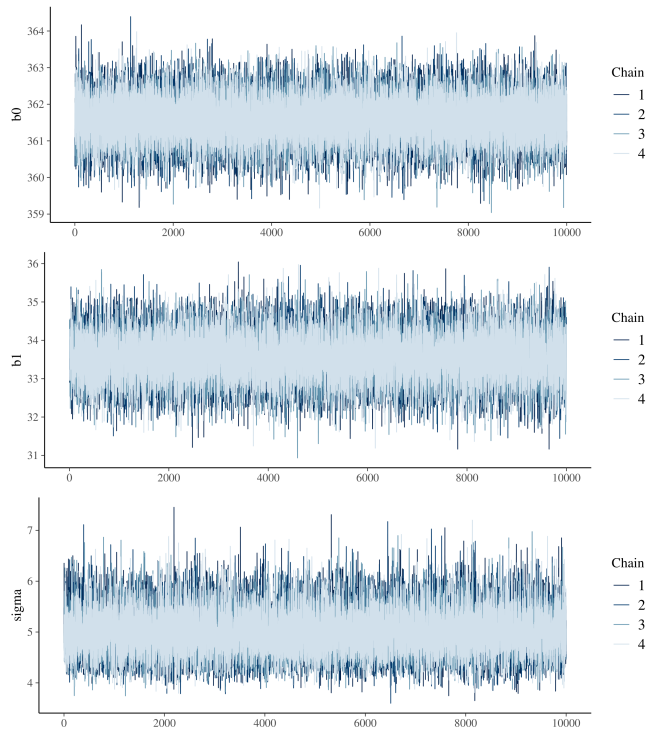


FIG. 12: Trace plots for the linear model parameters.

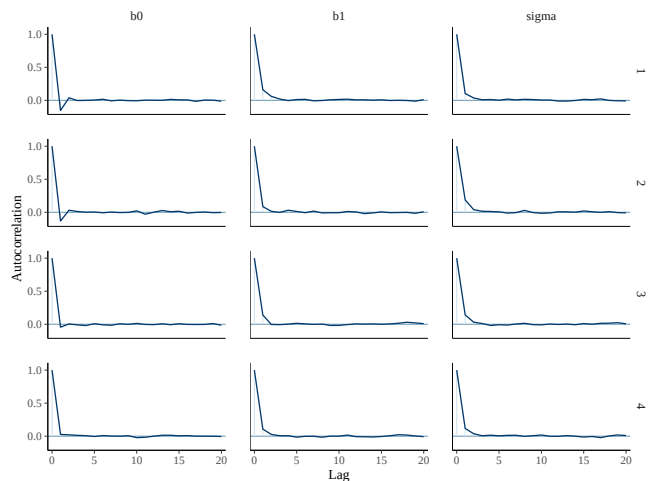


FIG. 13: Autocorrelation functions (ACF) for the linear model parameters.

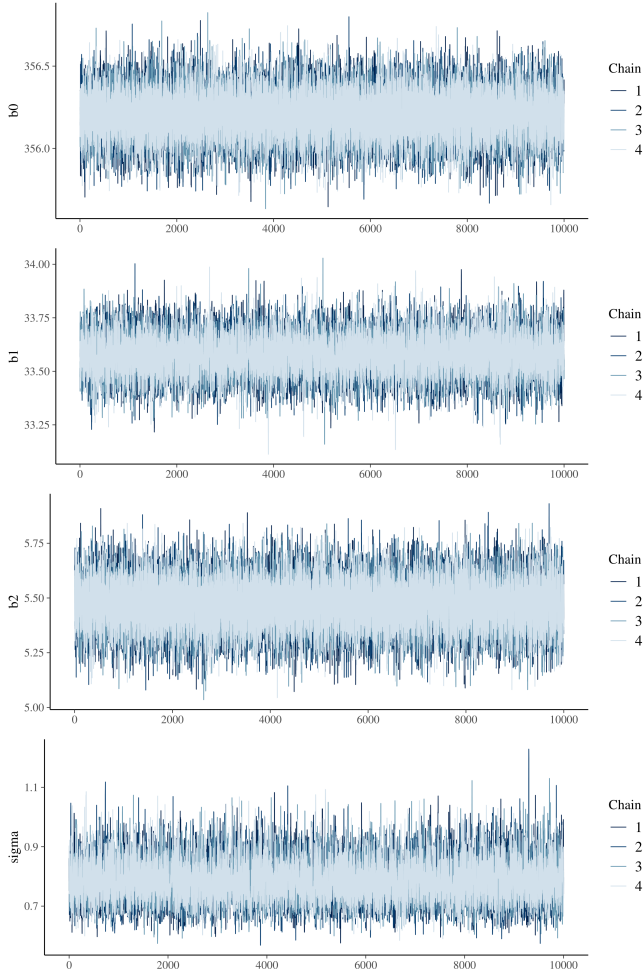


FIG. 14: Trace plots for the quadratic model parameters.

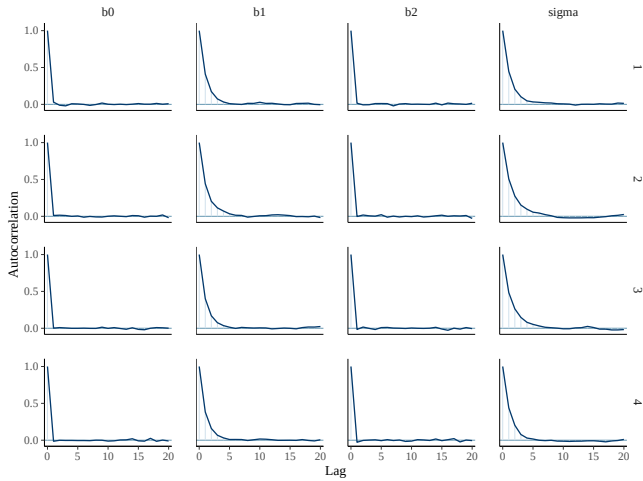


FIG. 15: Autocorrelation functions (ACF) for the quadratic model parameters.

CO₂ Growth Rate Distributions in the quadratic model

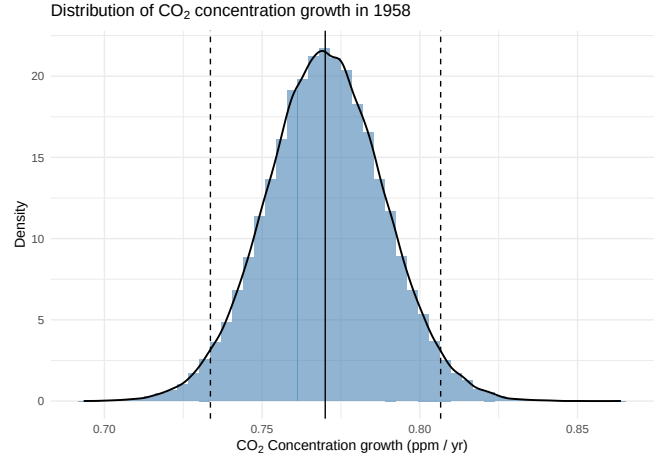


FIG. 16: Posterior distribution of the CO₂ growth rate for the 1958-1959 period, estimated using the quadratic model.

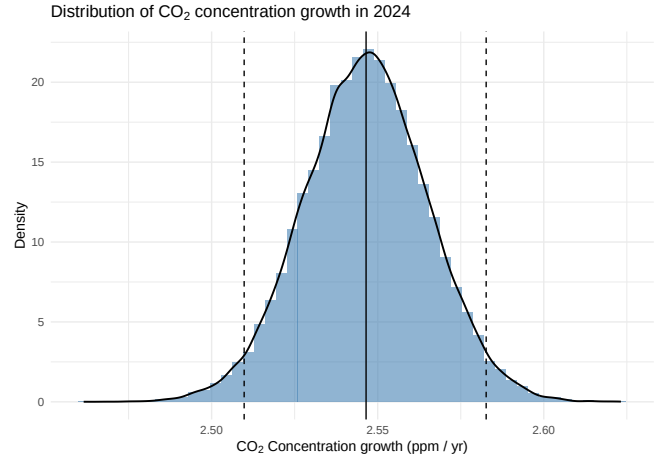


FIG. 17: Posterior distribution of the CO₂ growth rate for the 2024-2025 period, estimated using the quadratic model.

```

1 data {
2   int<lower=1> N;
3   int<lower=1> N_pred;
4   vector[N] t;
5   vector[N_pred] t_pred;
6   vector[N] y;
7 }
8 parameters {
9   real<lower=0, upper=1000> b0;
10  real<lower=-1.56, upper=1.56> alpha;
11  real<lower=-10, upper=5> logsigma;
12 }
13 transformed parameters {
14   vector[N] mu;
15   vector[N_pred] mu_pred;
16   real<lower=0> sigma;
17   real b1;
18
19   b1 = tan(alpha);
20   sigma = exp(logsigma);
21
22   mu = b0 + b1 * t;
23   mu_pred = b0 + b1 * t_pred;
24 }
25 model {
26   // Priors
27   b0 ~ uniform(0, 1000);
28   alpha ~ uniform(-1.56, 1.56);
29   logsigma ~ uniform(-10, 5);
30
31   // Likelihood
32   y ~ normal(mu, sigma);
33 }
34 generated quantities {
35   vector[N_pred] y_rep_pred;
36   vector[N] y_rep;
37   vector[N] log_lik;
38   for (i in 1:N) {
39     log_lik[i] = normal_lpdf(y[i] | mu[i],
40                             sigma);
41     y_rep[i] = normal_rng(mu[i], sigma);
42   }
43   for (i in 1:N_pred) {
44     y_rep_pred[i] = normal_rng(mu_pred[i],
45                                sigma);
46   }
47 }

```

```

1 data {
2   int<lower=1> N;
3   int<lower=1> N_pred;
4   vector[N] t;
5   vector[N_pred] t_pred;
6   vector[N] y;
7 }
8 parameters {
9   real<lower=0, upper=1000> b0;
10  real<lower=-1.56, upper=1.56> alpha;
11  real<lower=-10, upper=5> logsigma;
12  real<lower=-500, upper=500> b2;
13 }
14 transformed parameters {
15   vector[N] mu;
16   vector[N_pred] mu_pred;
17   real<lower=0> sigma;
18   real b1;
19
20   b1 = tan(alpha);
21
22   sigma = exp(logsigma);
23
24   mu = b0 + b1 * t + b2 * square(t);
25   mu_pred = b0 + b1 * t_pred + b2 * square(
26     t_pred);
27 }
28 model {
29   // Priors
30   b0 ~ uniform(0, 1000);
31   alpha ~ uniform(-1.56, 1.56);
32   logsigma ~ uniform(-10, 5);
33   b2 ~ uniform(-500, 500);
34
35   // Likelihood
36   y ~ normal(mu, sigma);
37 }
38 generated quantities {
39   vector[N_pred] y_rep_pred;
40   vector[N] y_rep;
41   vector[N] log_lik;
42   for (i in 1:N) {
43     log_lik[i] = normal_lpdf(y[i] | mu[i],
44       sigma);
45     y_rep[i] = normal_rng(mu[i], sigma);
46   }
47   for (i in 1:N_pred) {
48     y_rep_pred[i] = normal_rng(mu_pred[i],
49       sigma);
50   }
51 }

```

main.R

```

1 # ===== #
2 # PREPROCESSING #
3 # ===== #
4 library(dplyr)
5 library(ggplot2)
6 library(cmdstanr)
7 library(posterior)
8 library(bayesplot)
9 library(GGally)
10
11 df <- read.table("co2_mm_mlo.txt", comment.
12   char = "#", col.names = c(
13     "year",
14     "month",
15     "decimal_date",
16     "monthly_average",
17     "de_seasonalized",
18     "ndays",
19     "stdev",
20     "unc_monthly_mean"
21   ))
22 # Monthly CO2 plot
23 monthly_plot <- ggplot(df, aes(decimal_date
24   , monthly_average)) +
25   geom_line(linewidth = 0.3) +
26   geom_point(size = 0.5) +
27   labs(
28     x = "Year",
29     y = expression(CO[2] ~ "(ppm)"),
30     title = "Monthly CO2 measurements") +
31     theme_minimal()
32 ggsave("monthly_co2.pdf", monthly_plot,
33   width = 8,height = 5)
34
35 # Remove problematic months
36 df <- df %>%
37   filter(
38     !(year == 1984 & month == 4),
39     !(year == 1975 & month == 12),
40     !(year == 2026))
41
42 # Annual average
43 annual <- df %>%
44   group_by(year) %>%
45   summarise(
46     annual_mean =
47       if (first(year) <= 1974) {
48         mean(monthly_average)
49       } else {
50         weighted.mean(monthly_average, ndays)
51       },
52     .groups = "drop")
53
54 annual_plot <- ggplot(annual, aes(year,
55   annual_mean)) +
56   geom_point() +
57   labs(
58     x = "Year",

```

```

58   y = expression(CO[2] ~ "(ppm)"),
59   title = "Annual average CO2"
60 ) +
61   theme_minimal()
62
63 ggsave("annual_co2.pdf", annual_plot, width
64   = 8, height = 5)
65
66 print(as.data.frame(annual))
67 print(min(annual$annual_mean))
68 print(max(annual$annual_mean))
69 print(min(annual$year))
70 print(max(annual$year)-min(annual$year))
71
72 # ===== #
73 # STANDARDIZE TIME #
74 # ===== #
75
76 years = annual$year
77 years_mean <- mean(years)
78 print(years_mean)
79 years_sd <- sd(years)
80 print(years_sd)
81 t <- (years - years_mean) / years_sd
82 years_plot_grid = seq(min(years), max(years)
83   ) + 10, length.out = 1000)
84 t_plot_grid <- (years_plot_grid - years_
85   mean) / years_sd
86
87 stan_data <- list(
88   N = length(t),
89   N_pred = length(t_plot_grid),
90   t = t,
91   t_pred = t_plot_grid,
92   y = annual$annual_mean)
93
94 # ===== #
95 # COMMON MODEL FITTING FUNCTION #
96 # ===== #
97
98 analyze_MCMC <- function(fit, model, stan_
99   data, param_names, fit_name) {
100   # Compute statistics and plots for the
101   # specified MCMC fit and parameters
102
103   dir.create(fit_name, showWarnings = FALSE
104     )
105
106   draws <- fit$draws(param_names)
107   draws_df <- as_draws_df(draws)
108
109   # 1. Compute summary statistics
110   summary_stats <- posterior::summarise_
111     draws(
112       draws,
113       mean,
114       median,
115       sd,
116       ~posterior::quantile2(.x, probs = c
117         (0.025, 0.975)),
118       rhat,
119       ess_bulk,
120       ess_tail

```

```

114 )
115 print(summary_stats)
116
117 # 2. MAP estimate
118 init_fun <- function() {
119   inits <- list(
120     b0 = as.numeric(fit$summary("b0")$mean)
121     ,
122     alpha = as.numeric(fit$summary("alpha")$mean),
123     logsigma = as.numeric(fit$summary("logsigma")$mean)
124   )
125   #if the model is quadratic, then:
126   if ("b2" %in% fit$summary()$variable) {
127     inits$b2 <- as.numeric(fit$summary("b2")$mean)
128   }
129   return(inits)
130 }
131 map_fit <- model$optimize(
132   data = stan_data,
133   seed = 42,
134   init = init_fun
135 )
136
137 cat("\n=== MAP estimates & 95% CI ===\n")
138 for (param in param_names) {
139   map_val <- map_fit$summary(param)$
140     estimate
141   q2.5 <- summary_stats$q2.5[summary_stats$
142     variable == param]
143   q97.5 <- summary_stats$q97.5[summary_
144     stats$variable == param]
145
146   cat(sprintf("%s: %.3f (95% CI: [%.3f,
147     %.3f])\n", param, map_val, q2.5, q97
148     .5))
149 }
150
151 # 3. Trace plots
152 for (i in seq_along(param_names)) {
153   param <- param_names[i]
154   pdf(file.path(fit_name, paste0("trace_
155     plot_", fit_name, "_", param, ".pdf")
156     ), width = 8, height = 3)
157   print(mcmc_trace(draws, pars = param))
158   dev.off()
159 }
160
161 # 4. ACF plots
162 pdf(file.path(fit_name, paste0("acf_plots
163   _", fit_name, ".pdf")), width = 8,
164   height = 6)
165 print(mcmc_acf(draws, pars = param_names)
166 )
167 dev.off()
168
169 # 6. Pairs plots
170 corr_mat <- cor(as.matrix(draws_df[,
171   param_names]))
172 print(corr_mat)
173
174 my_diag <- function(data, mapping, ...) {
175   x <- GGally::eval_data_col(data, mapping$x)
176
177   q <- quantile(x, c(0.025, 0.975))
178   mu <- mean(x)
179
180   lines_df <- data.frame(
181     xintercept = c(mu, q),
182     type = c("Mean", "95% CI", "95% CI")
183   )
184
185   ggplot(data, mapping) +
186     geom_histogram(aes(y = after_stat(
187       density)), bins = 50, fill="
188       lightblue", color="black", alpha
189       =0.5) +
190     geom_density(aes(color="Density",
191       linetype="Density"), linewidth=1,
192       key_glyph = "path") +
193     geom_vline(data = lines_df, aes(
194       xintercept = xintercept,
195       color = type,
196       linetype = type
197     ), linewidth = 0.8 ) +
198     scale_color_manual(name = NULL, values
199       = c("Density" = "blue", "Mean" = "
200       red", "95% CI" = "green")) +
201     scale_linetype_manual(name = NULL,
202       values = c("Density" = "solid", "
203       Mean" = "solid", "95% CI" = "dashed
204       ")) +
205     theme_minimal() +
206     theme(legend.position = "none" # remove
207       legend)
208 }
209
210 p <- ggpairs(
211   draws_df[, param_names],
212   upper = list(continuous = wrap("density",
213     alpha = 0.7)),
214   lower = list(continuous = wrap("cor",
215     stars = FALSE, size = 5)),
216   diag = list(continuous = my_diag)
217 )
218
219 ggsave(
220   file.path(fit_name, paste0("pairs_", fit_
221     name, ".pdf")),
222   p,
223   width = 10,
224   height = 10
225 )
226 print(paste0("Plots saved in directory: "
227   , fit_name))
228 print(paste0("End of MCMC analysis"))
229 return(summary_stats)
230 }
231
232 # ===== #
233 # LINEAR MODEL FITTING #
234 # ===== #

```

```

210 linear_model <- cmdstan_model("linear_
      regression.stan")
211 linear_fit <- linear_model$sample(
212   data = stan_data,
213   chains = 4,
214   parallel_chains = 4,
215   iter_warmup = 2000,
216   iter_sampling = 10000,
217   seed = 42,
218   refresh = 0
219 )
220 linear_fit$cmdstan_diagnose()
221 linear_summary <- analyze_MCMC(
222   fit = linear_fit,
223   model = linear_model,
224   stan_data = stan_data,
225   param_names = c("b0", "b1", "sigma"),
226   fit_name = "linear"
227 )
228
229 # ===== #
230 #           QUADRATIC MODEL FITTING           #
231 # ===== #
232 quadratic_model <- cmdstan_model("quadratic
      _regression.stan")
233 quadratic_fit <- quadratic_model$sample(
234   data = stan_data,
235   chains = 4,
236   parallel_chains = 4,
237   iter_warmup = 2000,
238   iter_sampling = 10000,
239   seed = 42,
240   refresh = 0
241 )
242 quadratic_fit$cmdstan_diagnose()
243 quadratic_summary <- analyze_MCMC(
244   fit = quadratic_fit,
245   model = quadratic_model,
246   stan_data = stan_data,
247   param_names = c("b0", "b1", "b2", "sigma"
248   ),
249   fit_name = "quadratic"
250 )
251 # ===== #
252 #           MODEL COMPARISON                   #
253 # ===== #
254 # comparison using LOO
255 library(loo)
256 log_lik_linear <- linear_fit$draws("log_lik
      ")
257 log_lik_quadratic <- quadratic_fit$draws("
      log_lik")
258
259 # compute LOO for both models
260 loo_linear <- loo(log_lik_linear)
261 loo_quadratic <- loo(log_lik_quadratic)
262 comparison <- loo_compare(loo_linear, loo_
      quadratic)
263 print(comparison)
264
265 # ===== #
266 # REGRESSION CURVE AND POST PRED INTERVAL #
267 # ===== #
268 plot_posterior_predictive <- function(fit,
      model_name) {
269   # model_name must be either "linear" or "
      quadratic"
270
271   mu_mat <- as_draws_matrix(fit$draws("mu
      _pred"))
272   yrep_mat <- as_draws_matrix(fit$draws("y_
      rep_pred"))
273
274   mean_mu <- apply(mu_mat, 2, mean)
275   mu_lo <- apply(mu_mat, 2, quantile,
      0.025)
276   mu_hi <- apply(mu_mat, 2, quantile,
      0.975)
277
278   pred_lo <- apply(yrep_mat, 2, quantile,
      0.025)
279   pred_hi <- apply(yrep_mat, 2, quantile,
      0.975)
280
281   fit_df <- data.frame(
282     year = years_plot_grid,
283     mu = mean_mu,
284     mu_lo = mu_lo,
285     mu_hi = mu_hi,
286     pred_lo = pred_lo,
287     pred_hi = pred_hi
288   )
289
290   # Generate and save the plot
291   pdf_name <- paste0(model_name, "_fit_with
      _intervals.pdf")
292   pdf(pdf_name, width = 8, height = 5)
293
294   p <- ggplot() +
295     geom_ribbon(data = fit_df, aes(x = year,
      ymin = pred_lo, ymax = pred_hi, fill
      = "95% Prediction Interval"), alpha =
      0.2) +
296     geom_ribbon(data = fit_df, aes(x = year,
      ymin = mu_lo, ymax = mu_hi, fill = "
      95% Credible Interval"), alpha = 0.5)
      +
297     geom_line(data = fit_df, aes(x = year, y
      = mu, color = "Mean Trend"),
      linewidth = 0.5) +
298     geom_point(aes(x = years, y = annual$
      annual_mean), color = "black", size =
      1, alpha = 0.7) +
299     geom_vline(
300       xintercept = max(years),
301       linetype = "dashed",
302       color = "black",
303       alpha = 0.6
304     ) +
305     scale_fill_manual(name = NULL, values = c
      ("95% Credible Interval" = "red", "
      95% Prediction Interval" = "grey50"))
      +
306     scale_color_manual(name = NULL, values =
      c("Mean Trend" = "darkred")) +
307     scale_x_continuous(
308       breaks = pretty(fit_df$year, 10),

```



```

309     expand = expansion(mult = c(0.01, 0.05)
310   ) +
311   scale_y_continuous(
312     breaks = pretty(fit_df$mu, 8),
313     expand = expansion(mult = c(0.02, 0.05)
314   ) +
315   labs(
316     x = "Year",
317     y = expression(paste("C0" ["2"], " (ppm
318       )")),
319     title = paste("Posterior fit -", model_
320       name, "model")
321   ) +
322   theme_minimal() +
323   theme(
324     legend.position = "bottom",
325     text = element_text(size = 12),
326     axis.title = element_text(face = "bold"
327   ),
328     plot.title = element_text(face = "bold"
329   )
330 )
331 print(p)
332 dev.off()
333
334 cat(sprintf("Plot successfully saved to:
335   %s\n", pdf_name))
336
337 # ---- centered quantities at observed
338 # points ----
339 pdf_name <- paste0(model_name, "_centered
340   .pdf")
341 pdf(pdf_name, width = 8, height = 5)
342
343 center <- mean_mu
344 center_datapoints <- apply(as_draws_
345   matrix(fit$draws("mu")), 2, mean)
346 p <- ggplot() +
347   geom_ribbon(data = fit_df, aes(x = year,
348     ymin = pred_lo - center, ymax = pred_
349     hi - center, fill = "95% Prediction
350     Interval"), alpha = 0.2) +
351   geom_ribbon(data = fit_df, aes(x = year,
352     ymin = mu_lo - center, ymax = mu_hi -
353     center, fill = "95% Credible
354     Interval"), alpha = 0.5) +
355   geom_line(data = fit_df, aes(x = year, y
356     = 0, color = "Mean Trend"), linewidth
357     = 0.5) +
358   geom_point(aes(x = years, y = annual$
359     annual_mean - center_datapoints),
360     color = "black", size = 1, alpha =
361     0.7) +
362   geom_vline(
363     xintercept = max(years),
364     linetype = "dashed",
365     color = "black",
366     alpha = 0.6
367   ) +
368   scale_fill_manual(name = NULL, values = c
369     ("95% Credible Interval" = "red", "
370     95% Prediction Interval" = "grey50"))
371
372 +
373   scale_color_manual(name = NULL, values =
374     c("Mean Trend" = "darkred")) +
375   scale_x_continuous(
376     breaks = pretty(fit_df$year, 10),
377     expand = expansion(mult = c(0.01, 0.05)
378   ) +
379   scale_y_continuous(
380     breaks = pretty(c(pred_lo - center,
381     pred_hi - center), 8),
382     expand = expansion(mult = c(0.02, 0.05)
383   ) +
384   labs(
385     x = "Year",
386     y = expression(paste("C0" ["2"], " (ppm
387       )")),
388     title = paste("Posterior centered fit -
389       ", model_name, "model")
390   ) +
391   theme_minimal() +
392   theme(
393     legend.position = "bottom",
394     text = element_text(size = 12),
395     axis.title = element_text(face = "bold"
396   ),
397     plot.title = element_text(face = "bold"
398   )
399 )
400 print(p)
401 dev.off()
402
403 cat(sprintf("Plot successfully saved to:
404   %s\n", pdf_name))
405 }
406
407 # Linear model
408 plot_posterior_predictive(fit = linear_fit,
409   model_name = "linear")
410
411 # Quadratic model
412 plot_posterior_predictive(fit = quadratic_
413   fit, model_name = "quadratic")
414
415 draws <- as_draws_df(quadratic_fit$draws())
416
417 b0 <- draws$b0
418 b1 <- draws$b1
419 b2 <- draws$b2
420
421 # design matrix
422 future_years = seq(2026, 2035, by=1)
423 x <- (future_years - years_mean)/years_sd
424
425 # compute full posterior predictive mean
426 # curve
427 mu_mat <- sapply(x, function(t) {b0 + b1 *
428   t + b2 * t^2})
429
430 # posterior mean + CI
431 mean_mu <- apply(mu_mat, 2, mean)
432 mu_lo <- apply(mu_mat, 2, quantile,
433   0.025)

```

```

399 mu_hi <- apply(mu_mat, 2, quantile,
400               0.975)
401 future_forecast <- data.frame(
402   year = future_years,
403   mean = mean_mu,
404   ci_lower = mu_lo,
405   ci_upper = mu_hi)
406
407 future_forecast$mean <- round(future_
408   forecast$mean, 2)
409 future_forecast$ci_lower <- round(future_
410   forecast$ci_lower, 2)
411 future_forecast$ci_upper <- round(future_
412   forecast$ci_upper, 2)
413
414 cat("Quadratic Model: 10-Year Forecast (
415   Parameter-Based)\n")
416 print(future_forecast, row.names = FALSE)
417
418 dir.create("quadratic_posterior_true_means"
419   , showWarnings = FALSE)
420 for (i in seq_along(future_years)) {
421   mu_samples <- b0 + b1 * x[i] + b2 * x[i
422     ]^2
423
424   df <- data.frame(mu = mu_samples)
425
426   p <- ggplot(df, aes(x = mu)) +
427     geom_histogram(
428       aes(y = after_stat(density)),
429       bins = 50,
430       fill = "steelblue",
431       alpha = 0.6) +
432     geom_density(color='black', linewidth
433       =0.7) +
434     geom_vline(
435       xintercept = quantile(mu_samples, c
436         (0.025, 0.5, 0.975)),
437       linetype = c("dashed", "solid", "dashed
438         ")
439     ) +
440     labs(
441       title = paste("Posterior of True Mean
442         CO:", future_years[i]),
443       x = expression(paste("CO"[2], " (ppm)"),
444       y = "Density"
445     ) +
446     theme_minimal()
447
448     ggsave(
449       filename = file.path(
450         "quadratic_posterior_true_means",
451         paste0("posterior_mean-", future_years[
452           i], ".pdf")
453       ),
454       plot = p,
455       width = 7,
456       height = 5
457     )
458   }
459
460 # ===== #
461 #   DISTRIBS OF VELOCITY AND ACCELERATION   #
462 # ===== #
463
464 draws_lin <- as_draws_df(linear_fit$draws()
465   )
466 draws_quad <- as_draws_df(quadratic_fit$
467   draws())
468
469 t1 <- min(years)
470 t2 <- max(years) - 1
471
472 dist_lin_vel <- draws_lin[["b1"]] / years_
473   sd
474 dist_quad_v1 <- draws_quad[["b1"]]/years_
475   sd + draws_quad[["b2"]] * (1 + 2*(t1 -
476   years_mean))/years_sd^2
477 dist_quad_v2 <- draws_quad[["b1"]]/years_
478   sd + draws_quad[["b2"]] * (1 + 2*(t2 -
479   years_mean))/years_sd^2
480 dist_quad_acc <- 2 * draws_quad[["b2"]] /
481   years_sd^2
482
483 get_summary <- function(dist) c(mean = mean
484   (dist), quantile(dist, c(0.025, 0.975))
485   )
486
487 print(get_summary(dist_lin_vel))
488 print(get_summary(dist_quad_v1))
489 print(get_summary(dist_quad_v2))
490 print(get_summary(dist_quad_acc))
491
492 plots_data <- list(
493   velocity_linear = list(
494     data = dist_lin_vel,
495     title = expression(paste("Distribution of
496       ", CO[2], " concentration growth in
497       linear model")),
498     xlab = expression(paste(CO[2], "
499       Concentration growth (ppm / yr)"))
500   ),
501   velocity_1958 = list(
502     data = dist_quad_v1,
503     title = expression(paste("Distribution of
504       ", CO[2], " concentration growth in
505       1958")),
506     xlab = expression(paste(CO[2], "
507       Concentration growth (ppm / yr)"))
508   ),
509   velocity_2024 = list(
510     data = dist_quad_v2,
511     title = expression(paste("Distribution of
512       ", CO[2], " concentration growth in
513       2024")),
514     xlab = expression(paste(CO[2], "
515       Concentration growth (ppm / yr)"))
516   ),
517   acceleration = list(
518     data = dist_quad_acc,
519     title = expression(paste("Acceleration
520       Distribution of ", CO[2])),
521     xlab = expression(paste(CO[2], "
522       Acceleration (ppm / yr)^2, ")")
523   )
524 )

```

```

493 )
494
495 for (name in names(plots_data)) {
496   meta <- plots_data[[name]]
497
498   p <- ggplot(data.frame(value = meta$data)
499     , aes(x = value)) +
500     geom_histogram(
501       aes(y = after_stat(density)),
502       bins = 50, fill = "steelblue", alpha =
503         0.6) +
504     geom_density(color = 'black', linewidth =
505       0.7) +
506     geom_vline(
507       xintercept = quantile(meta$data, c
508         (0.025, 0.5, 0.975)),
509       linetype = c("dashed", "solid", "dashed"
510         ),
511       color = "black") +
512     labs(title = meta$title, x = meta$xlab, y
513       = "Density") +
514     theme_minimal()
515
516     ggsave(
517       filename = paste0(name, "_distribution.
518         pdf"),
519       plot = p,
520       width = 7,
521       height = 5
522     )
523 }
524
525 # ===== #
526 # POSTERIOR PREDICTIVE CHECKS #
527 # ===== #
528 # Linear Model
529 yrep_matrix_lin <- linear_fit$draws("y_rep"
530   , format = "draws_matrix")
531
532 p_ppc_lin <- ppc_dens_overlay(y = annual$
533   annual_mean, yrep = yrep_matrix_lin
534   [1:100, ]) +
535   labs(
536     title = "Posterior Predictive Check -
537       Linear Model",
538     x = expression(paste("CO"[2], " (ppm)")),
539     y = "Density") +
540     theme_minimal()
541
542 ggsave(
543   filename = "ppc_linear.pdf",
544   plot = p_ppc_lin,
545   width = 8,
546   height = 5)
547
548 # Quadratic Model
549 yrep_matrix_quad <- quadratic_fit$draws("y_
550   rep", format = "draws_matrix")
551
552 p_ppc_quad <- ppc_dens_overlay(y = annual$
553   annual_mean, yrep = yrep_matrix_quad
554   [1:100, ]) +
555   labs(
556     title = "Posterior Predictive Check -
557       Quadratic Model",
558     x = expression(paste("CO"[2], " (ppm)")),
559     y = "Density") +
560     theme_minimal()
561
562 ggsave(
563   filename = "ppc_quadratic.pdf",
564   plot = p_ppc_quad,
565   width = 8,
566   height = 5)

```

REFERENCES

- [1] J. G. Canadell, P. M. S. Monteiro, M. H. Costa, L. Cotrim da Cunha, P. M. Cox, A. V. Eliseev, S. Henson, M. Ishii, S. Jaccard, C. Koven, A. Lohila, P. K. Patra, S. Piao, J. Rogelj, S. Syampungani, S. Zaehle, and K. Zickfeld, in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2021) pp. 673–816.
- [2] X. Lan and D. C. Keeling, “Mauna loa atmospheric co2 monthly mean data,” (2026).
- [3] M. D. Hoffman and A. Gelman, “The no-u-turn sampler: Adaptively setting path lengths in hamiltonian monte carlo,” (2011), arXiv:1111.4246 [stat.CO].
- [4] A. Vehtari, A. Gelman, and J. Gabry, *Statistics and Computing* **27**, 1413–1432 (2016).